



TAKORADI TECHNICAL UNIVERSITY

COMPUTER SCIENCE DEPARTMENT

**BACHELOR OF TECHNOLOGY IN INFORMATION TECHNOLOGY –
DATA MANAGEMENT AND ANALYTICS**

GROUP 7

MINI PROJECT DOCUMENTATION ON E-COMMERCE WEBSITE

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1. Introduction

In today's digital world, online shopping has become a crucial part of everyday life. Our team developed an e-commerce platform as a mini project to provide a seamless shopping experience. The platform enables users to browse products, add items to their cart, and securely complete purchases.

The objective of this project was to create a user-friendly and efficient online store. It includes features like product search, secure payments, and order management, making it easy for both customers and store owners. This documentation provides a detailed overview of the project, including its design, features, technologies, and future improvements.

2. System Features

1. User Authentication

- Secure user registration and login system.
- Password reset and account recovery options.

User Login

Username

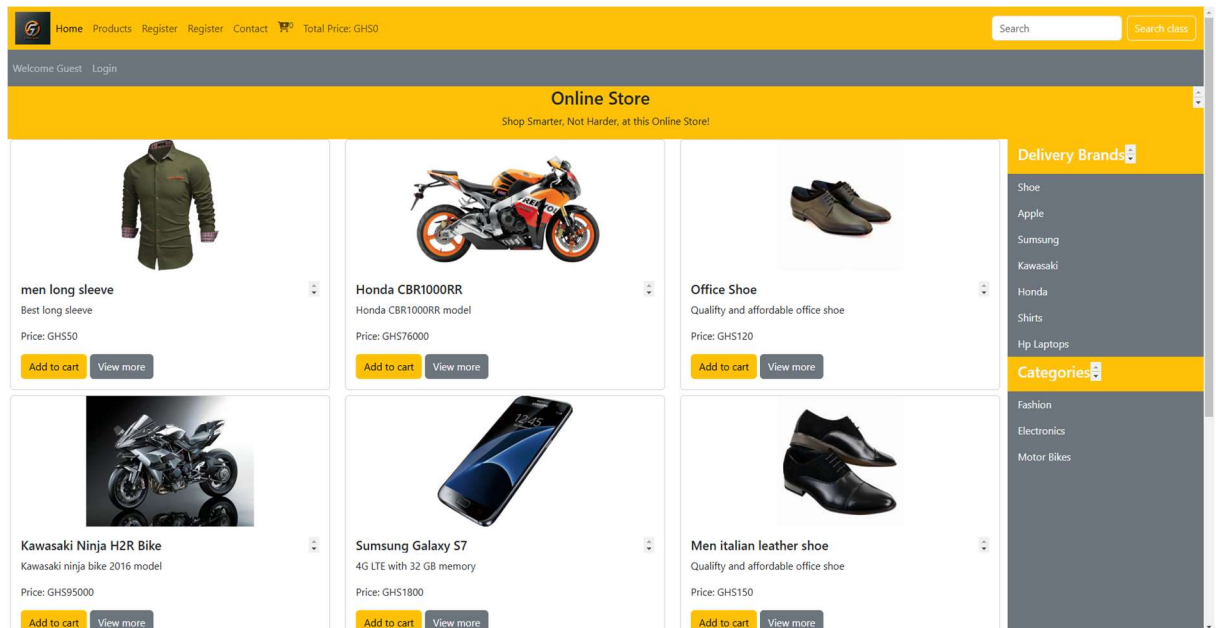
Password

Login

Don't have an account? [Register](#)

2. Product Catalog and Search

- Well-organized product listings with images and descriptions.
- Search functionality and category filters.



3. Shopping Cart and Checkout

- Add/remove products from the cart.
- Order summary and secure checkout process.

G-7 Online Store						
Shop Smarter, Not Harder, at G-7 Online Store!						
Product Title	Product Image	Quantity	Total Price	Remove	Operations	
Office Shoe			GH\$ 120	☐	<button>Update Cart</button>	<button>Remove Cart</button>
men long sleeve			GH\$ 50	☐	<button>Update Cart</button>	<button>Remove Cart</button>
Subtotal: GH\$170		<button>Continue Shopping</button>	<button>Checkout</button>			

4. Payment Gateway Integration

- Secure transactions using PayStack.
- Transaction verification for security.

5. Order Management

- Order tracking for users.
- Admin dashboard for managing orders.

All orders



Sl No	Due Amount	Invoice Number	Total Products	Order Date	Status	Delete
1	76000	1648166833	1	2025-03-03 20:33:04	pending	
2	170	518174707	2	2025-03-03 20:44:37	pending	

6. Security Features

- Secure authentication and encryption for user data.
- Protection against unauthorized access and fraud.

3. Technologies Used

1. Frontend Development

- Technologies: (HTML, CSS, JavaScript, etc.)
- Responsive design with Bootstrap.

2. Backend Development

- Technologies: (PHP)
- API endpoints for seamless communication.

3. Database Management

- Database: (MySQL, XAMPP).
- Secure storage of user, product, and transaction data.

4. Payment Integration

- Services: (PayStack)
- Secure encryption for payment transactions.

4. System Architecture

- **Three-tier architecture:**
 - Presentation Layer (Frontend)
 - Application Layer (Backend)
 - Data Layer (Database)
- **System Workflow:**
 1. User registers and logs in.
 2. Browses products and adds them to the cart.
 3. Proceeds to checkout and completes payment.
 4. Order details stored and managed in the database.
 5. Users track their orders, and admins manage transactions.

5. Implementation Details

1. Project Planning

- Defined project scope and assigned tasks.
- Created UI wireframes and database schema.

2. Development Process

- Frontend: Designed UI and integrated APIs.
- Backend: Developed authentication, product management, and order processing features.
- Database: Optimized queries and implemented security best practices.
- Payment Integration: Secure processing of transactions.

6. Testing and Evaluation

1. Unit Testing

- Tested individual functions such as login, cart, and checkout.

2. Integration Testing

- Verified API interactions and database transactions.

3. UI Testing

- Checked responsiveness and user interactions.

4. Performance Testing

- Optimized page load times and server responses.

5. Security Testing

- Prevented SQL injection and XSS attacks.
- Ensured secure authentication and encrypted payments.

7. Conclusion and Future Enhancements

Conclusion

This e-commerce platform successfully meets its objectives by providing an efficient and secure shopping experience. The project enhanced our knowledge of web development, database management, API integration, and security implementation.

Future Enhancements

- **AI-powered recommendations** to improve user experience.
- **Multi-vendor support** to allow different sellers to list products.
- **Mobile app version** for better accessibility.
- **Live chat support** for real-time customer assistance.
- **Real-time order tracking** with estimated delivery updates.
- **Loyalty programs** for customer retention.
- **Analytics dashboard** for store owners to monitor sales and trends.

Acknowledgments

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